SECURING CERTIFICATE MANAGEMENT SYSTEM USING HYPERLEDGER BASED PRIVATE BLOCKCHAIN

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A report submitted to the Department of Computer Science and Engineering Faculty, Bangladesh Army International University of Science and Technology in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering.

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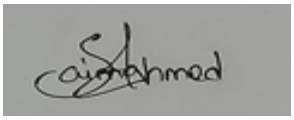
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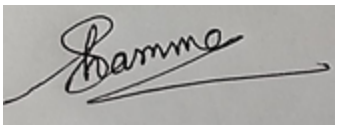
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ABSTRACT

Certificate digitalization has become a core part now-a-days. It includes information about certificates where data is stored digitally. So far, most of the certificates are not digital which are difficult to maintain, and not highly secured. For these reasons, forgery of certificates and duplication of certificates are continuously happening. Blockchain gives us the opportunity of storing data that are highly secured and more efficient to use. Every transaction in this system is immutable which means it is quite impossible to change the data or hack the system.In this paper, hyperledger based a low cost and private blockchain is used for accomplishing the certificate system. This system restricted the access of users in some way but the owner is allowed to access all the functions of managing certificates. This privileges more security,privacy and authentication. Transactions have been made easier to be updated shortly and insert the data into the blockchain. Deploying this blockchain based certificate system will be beneficial for the digital world as it will resolve the issues of maintenance and indemnity.

TABLE OF CONTENTS

DECLARATION I
APPROVAL II
ACKNOWLEDGEMENT IV
ABSTRACT V

CHAPTER 1 1
INTRODUCTION 1
1.1. Overview 1
1.2. Historical Background 2
1.3. Research Problem 3
1.4. Research Objective 3
1.5. Introduction to this Thesis 4
1.6. Future Scope of this study 4

CHAPTER 2 5
Literature Review 5
2.1. Overview 5
2.1.1. Bitcoin 5
2.1.2. Ethereum 6
2.2. Earlier Research 6

CHAPTER 3 9
System Overview 9
3.1. Proposed Approach 9
3.1.1. Blockchain 9
3.1.1.1 Why is blockchain famous? 10
3.1.1.2 Blockchain Prerequisites 11
3.1.2. Hyperledger Fabric 13
3.1.2.1. Why do we choose hyperledger fabric based blockchain? 14
3.1.2.2. Hyperledger Advantages 15
3.2. Proposed System Strategy 16

CHAPTER 4 19
Methodology 19
4.1. Required Tools 19
4.2. Technical Specification 25

CHAPTER 5 31
Result and Discussion 31
5.1. Experimental Result 31
5.2. Conclusions 38
REFERENCES 39

LIST OF FIGURES

Figure 2.1: BlockChain Transaction Method for Bitcoin 5

Figure 3.1: Blockchain Technology 10

Figure 3.2: Cryptography 12

Figure 3.3: Hyperledger fields 14

Figure 3.4: Resources of proposed system 16

Figure 3.5: System's Information Flow and Interaction with Blockchain of our proposed system architecture 17

Figure 4.1: Hyperledger Composer Playground (Layout 1) 21

Figure 4.2: Hyperledger Composer Playground (Layout 2) 22

Figure 4.3 : Hyperledger Composer Playground (Layout 3) 23

Figure 4.4: Hyperledger Composer Playground (Layout 4) 24

Figure 4.5: Transaction Operation for Implementing Creating and Updating Certificate 26

Figure 4.6: Asset Registry for certificate system 28

Figure 4.7: Authentication of Certificate Validity by Transaction Query 29

Figure 4.8: Access Control Limiting Blockchain Access 30

Table 5.1: Arbitrary student Information for system inspection purpose 31

Figure 5.1: Historian Record 32

Figure 5.2: Create Asset Transaction 33

Figure 5.3: Accessing from Historian Record for create asset transaction 34

Figure 5.4: Update Asset Transaction 35

Figure 5.5: Accessing from Historian Record for Update asset transaction 36

Table 5.2: Time Needed to Execute Transactions in Various Blockchain Technologies 37

Figure 5.6: Comparison of Computation Time for Three Blockchain Systems 37

Chapter 1

Introduction

1.1. Overview

Certificate is a valuable phenomenon in our life. The implications of certificates can not be denied. Certificate enriches opportunity of employment,accessing community involvement benefits and better opportunities for retaining a current job. The deficiency of maintaining a secure system can cause fabrication of certificates. Certificate represents one's ability and by observing it authority can justify about the person. In the case of a fabricated certificate, the justification will also be inaccurate. For this we need one kind of system where fabrication of certificates can be repelled. Blockchain has introduced us to build this kind of system where the management of certificates will be much more feasible and secured. Management of certificates mainly focuses on digitalization of certificates. When we alter the information of certificates into digital data, it is known as digitalization of certificates. Through digitalization we can prevent manipulated certificates, the illegal access of other's certificates.

The drawbacks of public blockchain is that anyone can see information which is not required and we can not control the access of users. The speed of public blockchain is relatively low. On the other hand ethereum requires transaction fees . Though ethereum is the second most decentralized cryptocurrency in the world after bitcoin,it has some issues. The issues include high traffic in the network,not fully immutable and some well known problems in the programming language of ethereum.In this paper, we used hyperledger blockchain which is an open source and private blockchain and everyone can not access every part. It has permissioned mode of operation and is used to build enterprise blockchains.

Hyperledger not only solves trust issues in the network but also ensures a direct approach through **peer-to-peer transmission** and **decentralization**. It's high level security, interoperable nature,modular design and high-end API (Application Programming Interface) support and cryptocurrency-agnostic structure makes it useful across different industries.

1.2. Historical Background

Certificates demonstrate one's capabilities, and authorities can verify a person's potential by looking at it.The certificate enhances career prospects, as well as access to community engagement perks and higher chances of keeping a present position.The failure of creating and maintaining a secure system might end in certificate fabrication.The foregoing explanation concludes that the ability to erase all traces of a certificate transaction makes the certificate system extremely insecure. The certificate system may be considered safe using Blockchain technology since it is impossible to change or eliminate any record of transactions from a block of Blockchain.

Hyperledger Fabric is a modularized blockchain architecture and application platform. Here, fabrics are implemented in a variety of sectors due to their modular design. The hyperledger fabric is utilized in the certificate system for the following reasons:

1) Hyperledger not only addresses network trust concerns, but it also provides a direct approach via peer-to-peer transmission and decentralization.For this decentralized system,it is much more applicable in certificate systems.

2) Certificates need a high degree of security, interoperability, modular architecture.These features are available in hyperledger fabric based blockchain.

3) The key characteristic of certificate digitalization should be high-end Application Program Interface (API) support, which hyperledger fabric provides.

4) This saves both time and effort.

1.3. Research Problem

Various methodologies have been proposed to implement the Ethereum-based certificate system. Some of these systems are decentralized, whereas others are built on bitcoins. There are a number of difficulties with using these in certificate systems:

1) Though digital infrastructure was developed to manage school credentials, this idea may be extended to a wide range of disciplines, including valid identification, professional qualifications, and so on.
2) Bitcoin has a systemic danger, as well as a lack of warning and insufficient functionality.
3) On the other hand, Ethereum is a public blockchain feature that allows anybody to examine the data included within the blocks.
4) It needs to be tested with more network situations, and the scheme's implementation in a blockchain network is also absent.
5) Moreover, misappropriation cannot be prevented, and forgery cannot be disputed.

1.4. Research Objectives

If the above mentioned flaws can be taken into account, an efficient certificate system can be developed that will benefit all. The proposed solution addresses these flaws in existing hyperledger based certificate systems in the following manner:

1) Building a certificate digitization system on a private hyperledger blockchain, where every transaction is immutable and not accessible to everyone.These solves lack of warning and insufficient functionality.
2) A unique asset ID is needed. As a result, any certificate with an asset ID is regarded as genuine which solves forgery and misappropriation.
3) As it is a permissioned network that puts limits on who has access to the network and what they can do with that as well. It requires the participation of a known participant.
4) Governance and access control in transactions is the main focus of the system.
5) Building a more secure system.

1.5. Introduction to this Thesis

This thesis is distributed in the following manner:

1) The chapter on Literature Review defines the related concepts and previously published works that are suitable to the thesis.
2) The process and efficiency of the proposed system are explained in the subsequent System Overview.
3) The Methodology chapter goes into the specifics of how the system is put in place and how it works.
4) Blockchain is represented by a set of programming languages, frameworks, and tools.
5) The output produced by the system is described and shown using pictures in the Experimental Results & Discussion chapter.
6) The final chapter, result and discussions,summarizes all aspects of the completed system and shows the experimental result.

1.6. Future Scope of this Study

It has a promising future in terms of research. For now, we're using Hyperledger Fabric, however hyperledger Sawtooth could be used in the future. Hyperledger Sawtooth is an open source enterprise blockchain-as-a-service platform that can run customized smart contracts without needing to know the underlying design of the core system. It provides an extremely modular and flexible platform for implementing transaction-based updates to shared state between untrusted parties coordinated by consensus algorithms. Sawtooth supports both serial and parallel transfer scheduling. Transaction scheduling and execution in Sawtooth reliably and effectively manage transactions that alter the same state addresses, including transactions within the same block. Hyperledger Fabric's concurrency management allows endorsers to perform transactions in parallel to maximize throughput, and each event is reviewed upon commit to guarantee that no other transaction has changed data it has read.

Chapter 2

Literature Review

2.1 Overview

2.1.1. Bitcoin

Bitcoin is a decentralized digital money that was first introduced in January of 2009. Bitcoin is classified as a cryptocurrency since it is protected by encryption. There are no real bitcoins; instead, balances are recorded on a public ledger that anyone can see although each record is encrypted. A vast amount of computational power is used to verify all bitcoin transactions, a process known as "mining." Bitcoin is not issued or backed by any banks or governments, and a single bitcoin has no monetary value. Bitcoin is still only accepted by a tiny number of internet businesses. As a result, relying solely on Bitcoins as a currency is unsustainable. It's also possible that governments will force retailers to stop accepting Bitcoins in order to track consumers' transactions. Furthermore, Bitcoin charges a transaction fee.

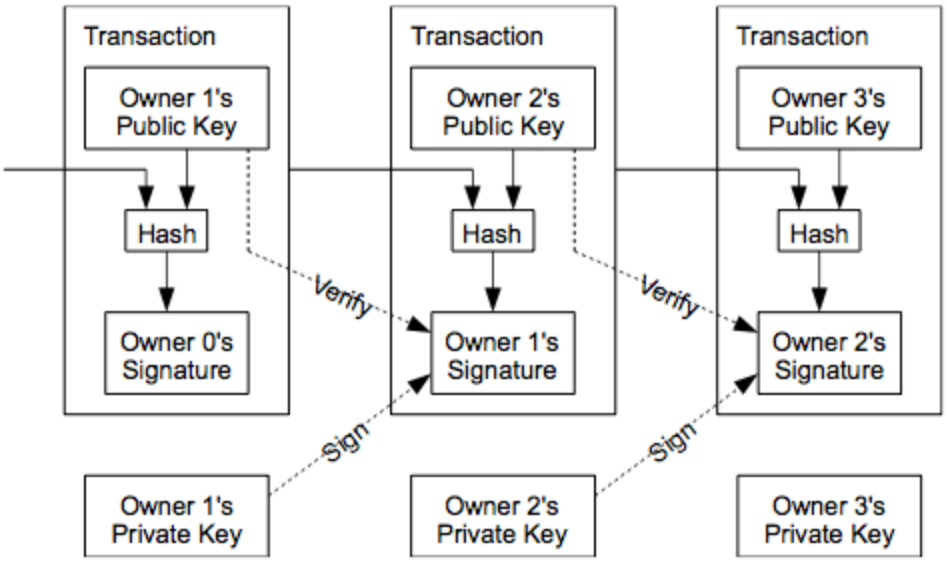


Figure 2.1: Blockchain Transaction Method for Bitcoin

2.1.2. Ethereum

Ethereum is a blockchain platform that has its own cryptocurrency, Ether (ETH), as well as its own programming language, Solidity. Ethereum is a decentralized public ledger for verifying and recording transactions as a blockchain network. Users of the network can create, publish, monetize, and use applications on the platform, and they can pay with Ether, the network's cryptocurrency. Insiders refer to the network's decentralized applications as "dApps." The users of Ethereum also need to pay transaction fees.

2.2 Earlier Research

Blockchain is a relatively new technology that is now being used in a variety of fields. In paper[1] the researchers proposed a hypothetical cloud and blockchain-based certificate verification system that could provide a potential solution for academic certificate issuing and verification in areas where cryptocurrencies are prohibited. In their paper, they didn't propose architecture for the graduate in the various stages such as undergraduate, postgraduate and others. And also, evaluation of performances of their architecture isn't presented in numerous parameters. In paper[2] other authors introduced the educational certificate blockchain system that can facilitate low latency and high throughput while also providing a means for query optimization. Though they used digital infrastructure to handle educational certificates, this theory can be applied to numerous fields such as proof of identity, proof of professional qualifications etc. In Paper[3] authors propose to secure the certificate by storing it in the blockchain system. The paper certificates must first be converted to digital certificates. The hash code value for the certificate is generated using the chaotic algorithm. certificates are stored in blockchain. The stored certificates in blockchain need to be validated by mobile application which will be troublesome in some aspects. For COVID-19 test takers, the researchers in paper [4] includes digital medical passports (DMP) and immunity certificates which demonstrate smart contracts based on the Ethereum blockchain that have been basically developed and tested to retain a digital medical identity, allowing for a quick and trusted response from the adequate medical authorities. They utilized an ethereum-based blockchain, which is risky and can lead to vulnerabilities, breakdowns, hacks, and other issues. In paper [5] their research proposes to

harness the power of distributed caching and investigates the feasibility of storing validated certificates in the cache areas of all IoT devices as a huge pool. In paper[6] they propose and implement the UniCert issuing and verification methodology, which is based on UniCoin. Their diagram can be used to solve a variety of additional concerns, including anti-counterfeiting, music product copyright protection, patents, and so on. As their work is based on Unicoin which is a digital money, that means it involves cryptocurrency. The authors of paper [7] construct an examination of blockchain adoption in relation to the process of creating meaningful certificates for higher education students. In their proposed DASC(dApp for Smart Certificates),they didn't show the evaluation and validation of uses of their desired solution. In addition,the application of this project in the higher education sector is also missing. The authors of paper[8] described a blockchain-based system and its software architecture that might offer a tamper-proof system allowing public users to validate the authenticity and status of ePhyto certificates to the growing ePhyto certificate system. Paper [9] is about to provide a server to store job training documents and certificates based on the blockchain platform open to the public. Because IPFS(Inter Planetary File System) is constantly growing and adopted by a lot of users, the storage issue is not solved. Furthermore,the trust issue also remains here though data is stored in a distributed and transparent environment of blockchain. In paper[10] the authors presented a certificate management platform for performance evaluation during recruitment using Bitcoin support and Ethereum. There is systemic risk in bitcoin, as well as a lack of warning along with inadequate functions. Ethereum, on the other hand, is a public blockchain feature that enables anybody to view the data contained within the blocks. More structured blockchain-based authentication and secure certificate revocation scheme is submitted by the authors of paper[11] which mainly constructed on the the signature module of X.509 digital certificate by replacing certificate hash value and a consensus algorithm is given for better efficiency. On the proposed algorithm,they need to check it with more network scenarios and implementation of the scheme in blockchain network based is also missing. Applying smart contracts for development to better the efficiency is also needed. The authors in paper[12] created BCASVADN, a blockchain-enabled certificate-based authentication mechanism for ITS car accident detection and notification. Through the authentication process in BCAS-VADN, each vehicle can securely notify its

adjacent Cluster Head of accident-related transactions whenever an accident is observed on the road by its own or neighboring vehicles. Here, the Road-Side Units (RSU) and the Edge Servers (ES) take place via wireless communication and the Internet and are also authenticated by blockchain which makes the system time consuming and more complex. Moreover, internet issues can also hamper this system. In paper[13] authors used blockchain and smart contracts for digital certificates. However,the authors of this paper made their certificate system depending upon ethereum and the system's application was programmed on the Ethereum platform and is run by the EVM as well as they have also used QR code here. As a result, the system becomes considerably more complicated, and using Ethereum transactions is much slower in this system, as transactions are only three or four per second. However, in order to digitize certificates, the number of transactions per second must be increased. Paper[14] is about copyright certificate storage and trading systems based on blockchain. This study investigates and examines the benefits of blockchain in copyright protection, as well as the present state of research, designing a copyright certificate storage and trading system based on blockchain. Though their system is much more reliable for fabric,the whole system's working process is time consuming because of many kinds of transaction queries and it becomes much more complex. The authors of paper[15] researched a cryptographic accumulator based distributed certificate revocation list. They focused on the accurate and efficient delivery of certificate revocation information in their study. They provided a design for an effective certificate revocation system based on an asymmetric cryptographic accumulator. The proposed system has limitations which include that certificate owners can not submit their revocation request to the revocation system blockchain for reducing the window of vulnerability that may be caused by the CA / VA update delay. In paper[16] the authors researched a blockchain system for e-learning assessment and certification which encompasses an entirely new network structure based on the combination of public and private blockchains, as well as four distinct smart contract schemes for e-learning assessment and credit exchange, digital certificate issuance and secure storage, digital certificate verification, and e-learning voucher allocation. The proposed approach has been shown to be a viable contender for creating a more equitable, healthy, and open e-learning and online education atmosphere. But the implementation of smart contracts for each functional module is missing in this system.

Chapter 3

System Overview

3.1 Proposed Approach

We implemented a certificate digitization system using a private hyperledger blockchain where every transaction is immutable and can not be accessed by everyone. Admin or owner has the ability to create a certificate asset and users have some restrictions while using the system. Every certificate asset has an asset ID by which that can be uniquely identified. So those certificates which have an asset ID are considered as a real one. Necessary information will be collected from the user. If there is some mistake or errors in the required information,it can be reformed. All transactions will be stored which can not be changed or modified. That means,if some information gets some errors and if the information is corrected later; both the error and the updated transactions will be stored. There is also a transaction which is called 'view transactions' where if anybody sees the certificate, the information will be stored in transactions. For this,illegal access of certificates can be easily determined and thus the security is very tight here.

3.1.1. Blockchain

A blockchain is a growing list of documents, known as blocks, that are cryptographically linked together. A cryptographic hash of the preceding block, a timestamp, and transaction data are all included in each block (generally represented as a Merkle tree). To get into the hash, the timestamp validates that the transaction data existed when the block was published. Because each block contains information about the one before it, they form a chain, with each new block reinforcing the preceding ones. As a result, blockchains are resistant to data tampering since the data in any one block cannot be changed retrospectively without affecting all subsequent blocks. Blockchains are often administered via a peer-to-peer network for use as a publicly distributed ledger, with nodes communicating and validating new blocks using a protocol. This type of design promotes a stable workflow in which participants' concerns about data security are minimal. The implementation of a blockchain removes a digital asset's attribute of limitless

reproduction. It verifies that each unit of value was only transferred once, putting an end to the long-standing issue of double spending. A value-exchange protocol has been described as a blockchain. When correctly set up to explain the trade agreement, a blockchain can guarantee title rights by providing a record that compels offer and acceptance.

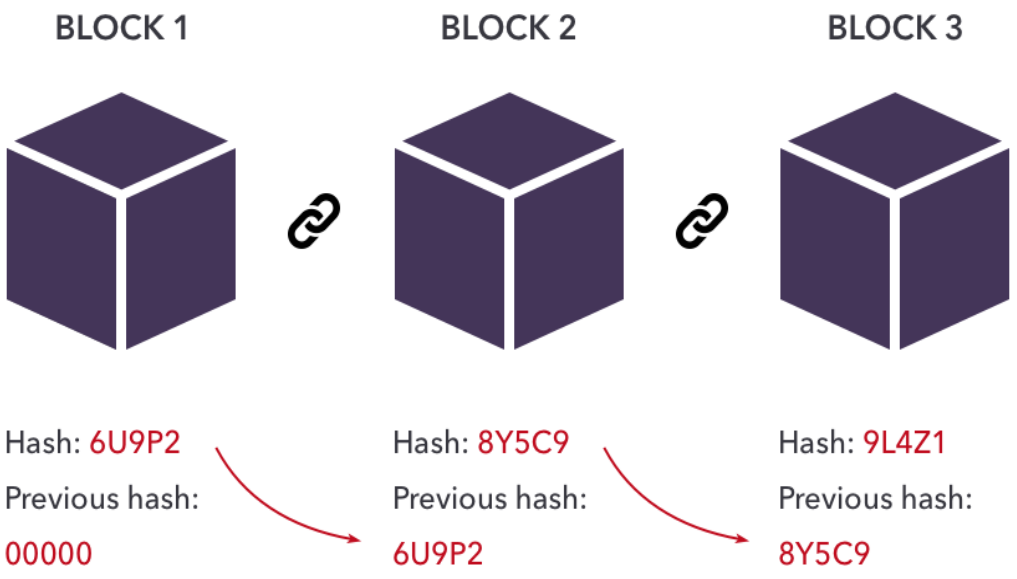


Figure 3.1: Blockchain Technology

3.1.1.1 Why is blockchain famous?

Blockchain is known for certain reasons.They are given below:-

1. **Open source:** In today's world,internet is one of the most important thing in our life.We can do business,use several sites like facebook,twitter,amazon,daraz etc.It has make our life more easier.But it seems that internet is not open source for everyone.It is controlled by few companies which we don't want.Blockchain gives us the feature of open source where anybody from anywhere can access and do transactions.

2. **Solved Money issue:** If people want to send money,it is given by bank or any kind of transactions.But the process and charge fee is much costly for transactions.For example,in Paypal 4% money is charged from transaction which is costly.In this case,blockchain have such no issues.
3. **Time:** General transaction takes much time whereas blockchain uses decentralized servers where the speed of transaction is faster.
4. **Trust:** In blockchain,every transaction is stored in a ledger and that ledger is transparent.So if anyone wants to misinterpret it,that's not possible.
5. **Amutibilty:** Transactions should not be changed.In blockchain as previously said,transactions are stored in ledger which is not changeable and remains for a long time.It indicates the immutable feature of blockchain.
6. **Safe:** The entire blockchain system works on cryptography.That's how it becomes more secure.

3.1.1.2 Blockchain Prerequisites

There are not many prerequisites for working in blockchain.But there's a twist.The twist is,it can be easy or difficult for people depending upon what they know.However, there are many things that needs to considered while getting into blockchain industry.They are given below:

1. **Distributive Computing:** Computing is used at the distributed invention stage of blockchain technology to create decentralized apps and information storage.It allows data to be exchanged and an open record database to be maintained in the executive system. The ledger may be defined as a decentralized ledger of any transactions or contracts maintained over many geographical connections made up of network nodes. Each contender at each network node has access to the network's registered data and that might control an identical copy of it.
2. **Cryptography:** Blockchain works on peer-to-peer(P2P) networks which simply means if there are multiple nodes,we don't need a central server.Like,torrent server.However,there are some concerns regarding it.They are-

1. *Confidentiality:* If some nodes send confidential messages to other nodes,that should be confidential.That means,it should not be seen by other nodes in that network.
2. *Integrity:* The transaction or the messages should not be modified or be changed.
3. *Non repudiation:* This term simply means the proof of transaction.
4. *Authentication:* Transactions or messages can be sent to someone else's behalf which we don't want.That means ,authentication is needed.

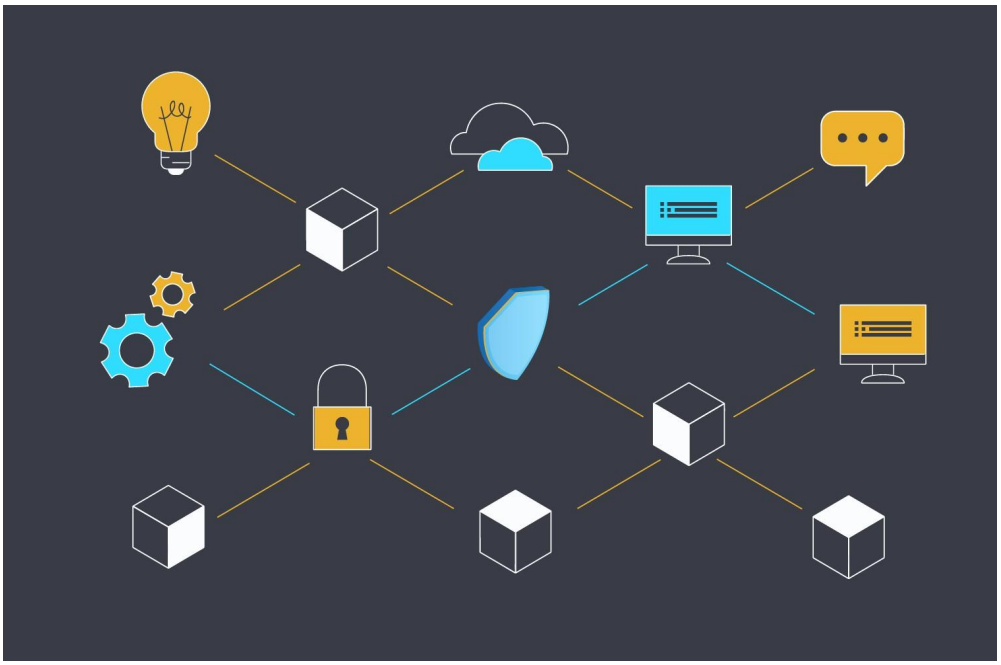


Figure 3.2: Cryptography

However, the word cryptography is divided into two parts.Crypt means hidden/secret and graphy means writing.It uses the concept of encryption and decryption. For that a hash key is used.

3. **Game theory:** Many features of blockchain (for example, the consensus process) may be built from a game theory framework. Rational smart contracts, game theoretic tactics, and rational mining techniques are all examples of game theory and blockchain linked.

3.1.2. Hyperledger Fabric

Hyperledger Fabric is designed to form the basis for creating modular applications and solutions. Plug-and-play components, such as consensus and membership services, are achievable with Hyperledger Fabric. It's modular and adaptable architecture provides a wide range of industry applications. It takes a simple technique to reach consensus that permits for scalability while maintaining privacy.It is a global collaboration,which is hosted by the Linux Foundation.It is like a greenhouse for blockchain. Hyperledger is not a blockchain nor a cryptocurrency. It's a project. In hyperledger, there are frameworks and tools. The frameworks are-

1. Iroha
2. Sawtooth
3. Fabric
4. Indy
5. Burrow

Fabric and sawtooth are the most famous frameworks.We have also tools available which is-

1. Hyperledger Composer
2. Hyperledger Cello
3. Hyperledger Quilt
4. Hyperledger Explorer
5. Hyperledger Caliper

These frameworks and tools are used to make blockchain. For more specifically, it is mainly used for permissioned blockchain, but we can also view them from public blockchain.

3.1.2.1 Why do we choose hyperledger fabric based blockchain?

Hyperledger have introduced us to private blockchain.There are certain reasons for choosing hyperledger fabric which is mentioned below:

1) Bitcoin and ethereum are both public blockchain which means anyone from anywhere can see the transactions whereas hyperledger is not public.
2) Blockchain is not just for cryptocurrencies. There are many fields like finance industry,insurance industry,education industry where blockchain consensus is needed.

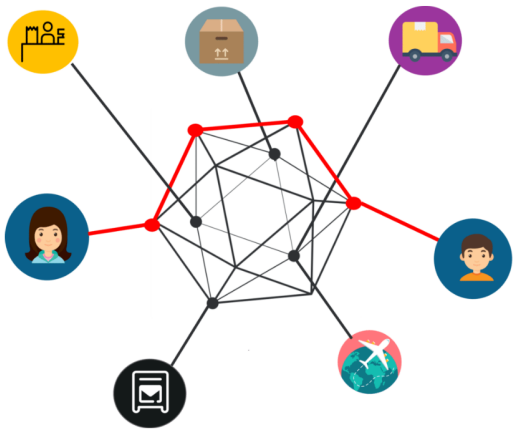


Figure 3.3: Hyperledger fields

3) In public blockchain,the data is transparent, which means the data is visible to everyone in that network.Anyone from anywhere can join the network just by installing a software or downloading the entire ledger.But the new node can be malicious.There's no way to finding that the new malicious node is not corrupting the data which is why consensus algorithm is needed.
4) Ethereum is the second most decentralized cryptocurrency in the world after bitcoin but it requires transaction fees.On the other hand,hyperledger does not require transaction fees and is easy to operate.
5) The transactions of public blockchain require more time and huge computer power.Because the complexity depends on network size,active users,blockchain size etc.

3.1.2.2 Hyperledger Advantages

Hyperledger's application field is steadily expanding due to its benefits.They are mentioned below:

1) As previously discussed,hyperledger is both an **open source and permissioned system** where anybody from that network can access the network by authentication.
2) Hyperledger gives the privileges for **governance** of the system by admin and easy access control for the end users. Admins have more privileges than the end users.
3) Decentralized networks are substantially more dispersed than centralized networks. Each node in the network operates as its own authority, with its own decision-making authority over how it interacts with other systems. These networks also share processing power and task functions among the servers that are linked.As hyperledger is a **decentralized network**,it ensures that a transaction can't be altered in any way.
4) The Hyperledger Fabric can initially handle 3,000 **transactions** per second, but with certain architectural tweaks, it can handle up to 20,000 per second.
5) System is more stable and more reliable for it's faster speed and permissioned system.
6) Transactions on Hyperledger do not involve the use of cryptocurrency. It lacks a built-in native coin, like ethereum's Ether. As a result, there is **no need for mining**. This makes it possible to create a scalable consensus algorithm that can accommodate the high transaction rates required by most corporate applications.
7) Hyperledger Provide **transparency and traceability of data shared across a network**. Nodes can select between No-op (no consensus required) and an agreement protocol (PBFT), which allows two or more parties to agree on a key while influencing the outcome. This prevents unwanted third parties from influencing the parties to make a crucial decision. As a result, Hyperledger provides fine-grained control over consensus and limited transaction access, resulting in increased efficiency, scalability, and privacy.
8) The transactions and data transmitted between two nodes are kept private in this system. Hyperledger is distinguished from other systems by the fact that dara and transactions are not shared with other nodes in the network.

3.2 Proposed System Strategy

In our certificate digitalization system,the admin or owner has the ability to create a certificate asset and users have some restrictions while using the system. This is because it is built on a hyperledger based blockchain.The following figure illustrates the design of our system.

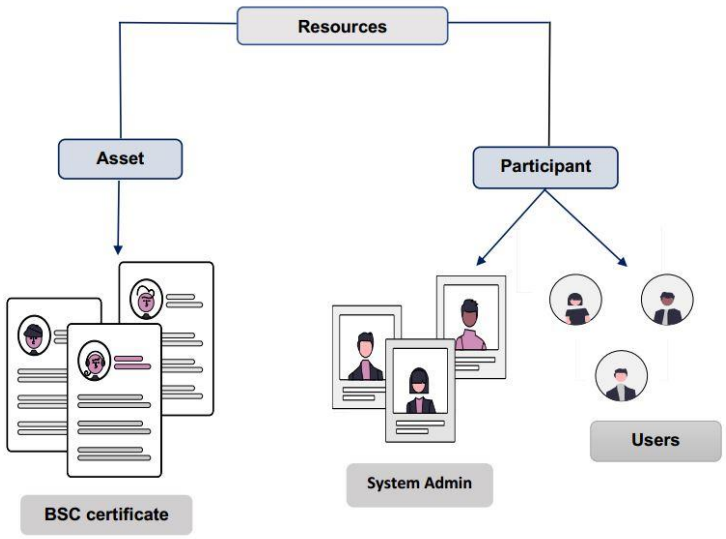


Figure 3.4: Resources of proposed system

The components that supply the system's intrinsic capabilities and contribute to its overall performance are termed as system resources.System admin and user are defined as participants of our certificate system.Admin collects necessary information from users and he have the ability of creating certificate asset named as BSC certificate whereas user have constraints regarding it.

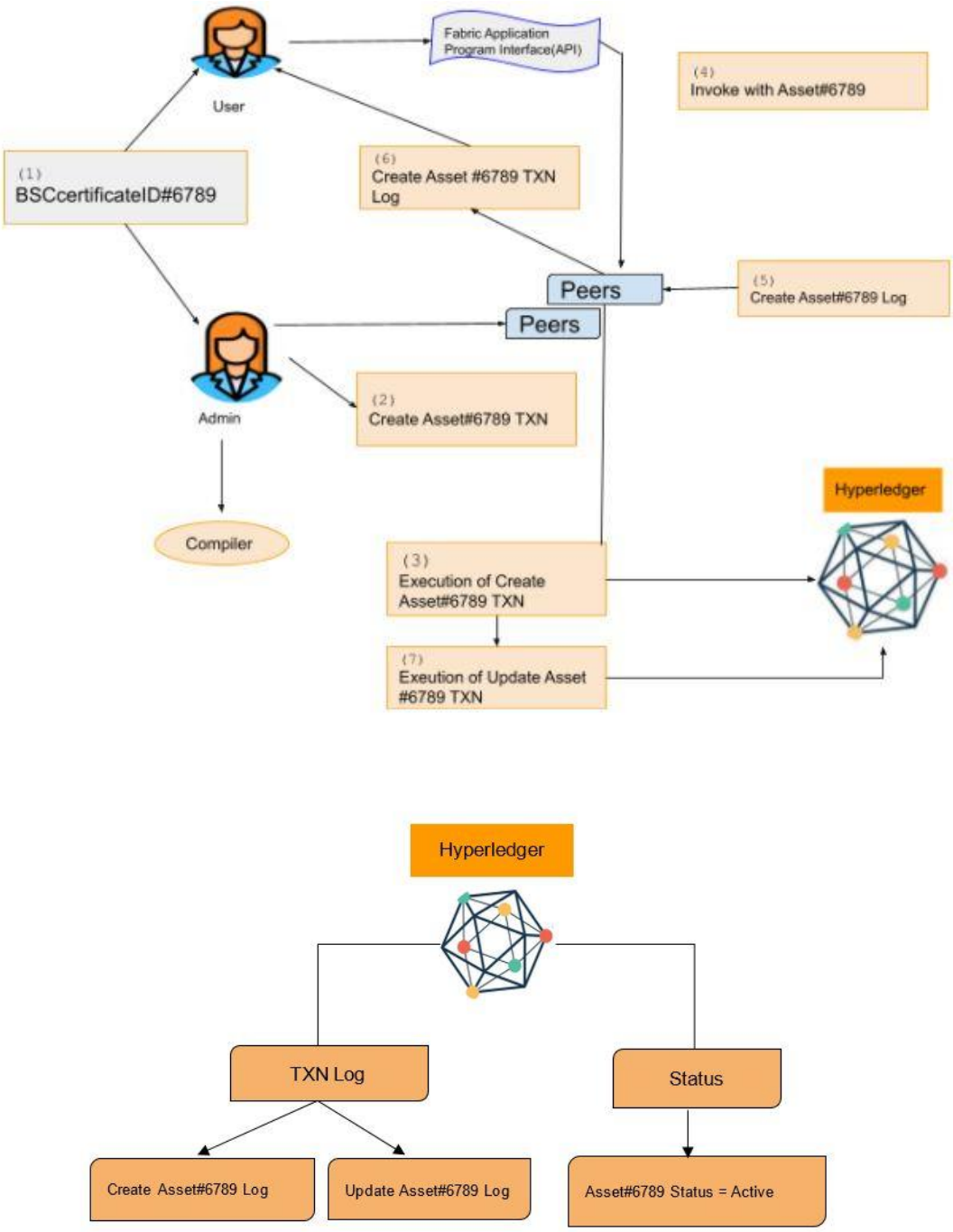


Figure 3.5: System's Information Flow and Interaction with Blockchain of our proposed system architecture

Asset ID will be shared by both admin and user. Using the required information,the block's asset is created by the admin and then the block generates a hash ID.With the hash ID,a create transaction will be performed and the blocks will be added to the system as a chain. However,this network is composed primarily of a set of peer nodes. Peers are essentially a network's most basic component. A chaincode contains the Hyperledger Fabric. Smart contracts and ledgers, respectively, are used to encapsulate shared processes and information in a network. These characteristics of a peer make them an excellent starting point for learning about a Fabric network. Through the peers,the system admin can check the asset registry and then he can update the asset information. It also generates the update transaction in the hyperledger. The fabric API(Application Program Interface) can be used by a user to access his own data.

Create transaction and update transaction event happened by the following steps:

1) Firstly,when a user wants to go into the certificate system,an ID must be provided. This certificateID will be available to both the system admin and user.
2) Admin then takes necessary information for creating certificates and this data is being compiled and saved in the system for further process.
3) In this step an TXN log is created for asset creation or we can say for ‘create transaction’ process.TXn log basically means transaction log.It keeps the data or history of actions for managing the system.
4) Finally, execution of create assets is done by hyperledger where transaction log and the status of transaction is kept.Status of transaction means if the asset created is active or inactive.
5) However,after successfully the data is being kept in hyperledger,users can also see it through peers.
6) For update transaction,the procedure is same as create transaction.Update transaction is done through both peers and fabric API ,and goes to hyperledger.Then a TXN log is created for update asset.

Chapter 4

Methodology

4.1. Required Tools

1. Install Git-2.28.0-64 bit:

   Git for Windows focuses on providing a lightweight, native collection of tools that bring the Git SCM's full feature set to Windows while providing adequate user interfaces for both expert and novice Git users.

2. Install Node JS:

   Node.js is an open-source, cross-platform framework that enables the building of fast web servers in JavaScript using event-driven programming. Utilizing a simplified approach of event-driven programming that employs callbacks to mark the end of a task, developers can design scalable servers without using threading.

3. Install Python 2.7.17

   Python is created under an OSI-approved open source license, which allows it to be freely used and distributed, including for commercial purposes. Thousands of third-party Python modules can be found in the Python Package Index. Python's standard library, as well as community-contributed modules, open up a world of possibilities.

4. Set-ExecutionPolicy AllSigned

   Unsigned scripts obtained from the internet, such as email and instant messaging programs, will not be launched by PowerShell. Without modifying the execution policy, we can sign the script or choose to run an unsigned script.

5. Install choco python visual cpp-build-tools -y

   By encapsulating installers, executables, zips, and/or scripts into a built package file, Chocolatey packages contain everything needed to handle a specific piece of

software into a single deployment artifact. MSBuild is a build tool that aids in the automation of the software development process, including source code compilation, packaging, testing, deployment, and documentation creation. It is possible to create Visual Studio projects and solutions without having the Visual Studio IDE installed using MSBuild.

After the installation of chocolatey packages and visual cpp build tools the following commands have to run-

- npm config set msvs_version 2017
- npm install -g composer-cli@0.19
- npm install -g composer-rest-server@0.19
- npm install -g generator-hyperledger-composer@0.19
- npm install -g yo
- npm install -g composer-playground@0.19

Lastly run Using the Command: composer-playground

6. Hyperledger Composer Playground

The Hyperledger Composer Playground is a user interface for deploying, configuring, and testing a business network. Users may maintain the security of the business network, invite participants to business networks, and link to various blockchain business networks using advanced Playground features. Under the Hyperledger initiative, Hyperledger composer is a suite of business-oriented tools for developing enterprise blockchain applications. Composer was designed to make it easier to create open-source blockchain applications that encouraged collaboration within companies and across networks.

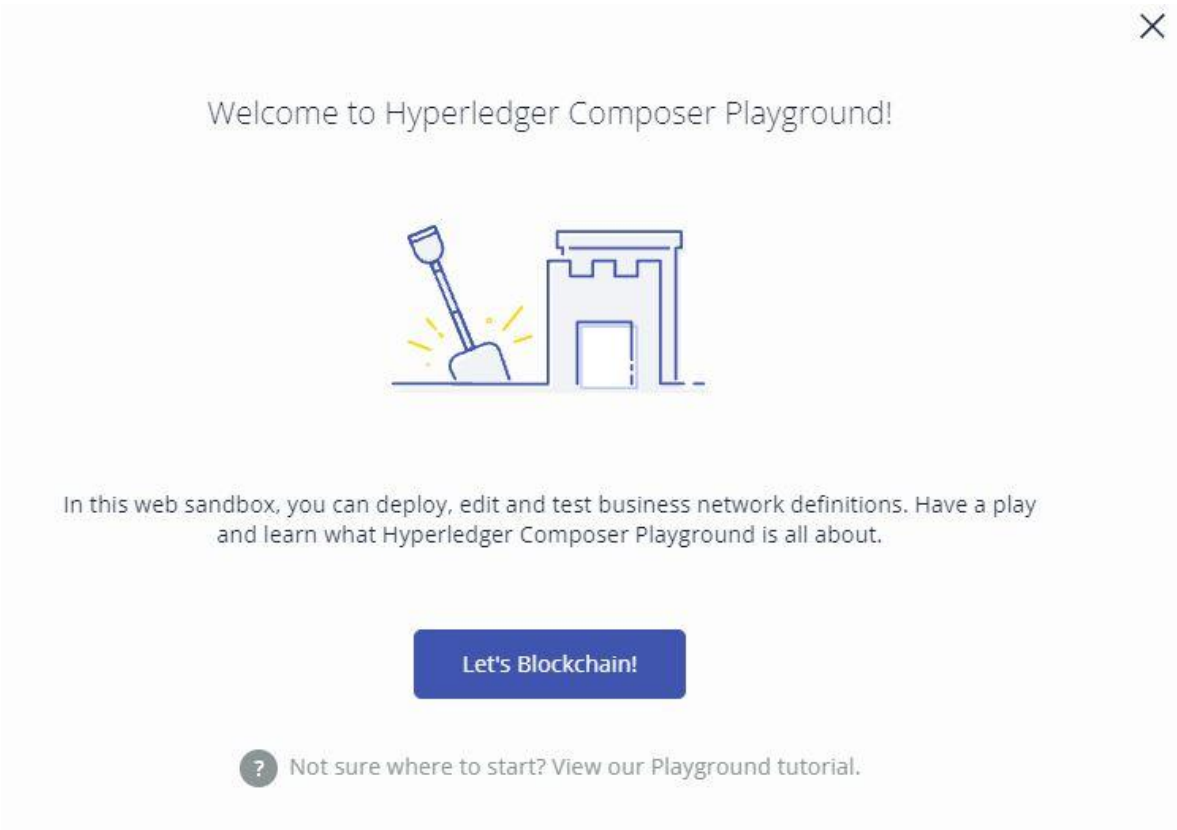


Figure 4.1: Hyperledger Composer Playground (Layout 1)

Figure 4.1 shows the playground of hyperledger composer which is the first layout of the playground.Whenever anybody runs the command-*composer-playground* in the administration panel or git bash administration panel then this layout appears.

After hit the button " Let's Blockchain" we will be able to see the next layout which is given below-

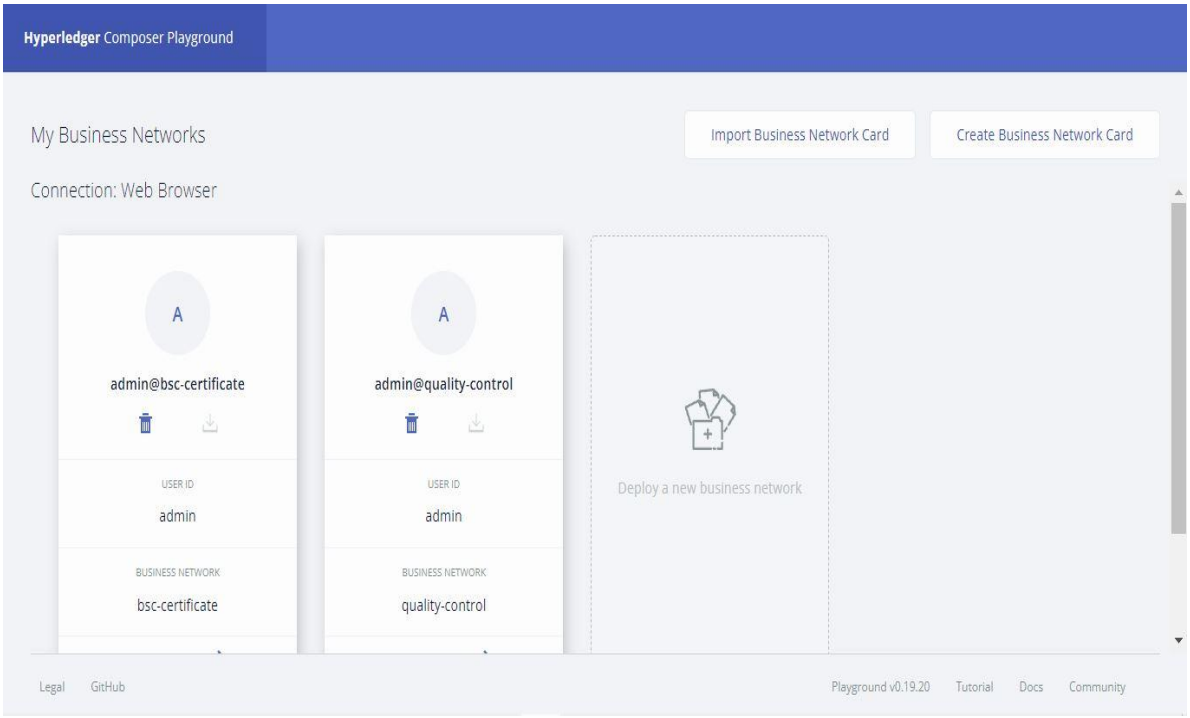


Figure 4.2: Hyperledger Composer Playground (Layout 2)

Here, business network cards for different kinds of systems are shown.

"Deploy a new business network" card basically means creating a new network in the blockchain hyperledger system. Figure 4.3 shows the basic information for creating a new business network. The basic information needed to deploy a network are, name of the network,the use of the business network and the name of a network admin card. These things needed to be filled out correctly as shown in the layout. Otherwise the network will not be deployed.However,some templates are also given for further procedure.

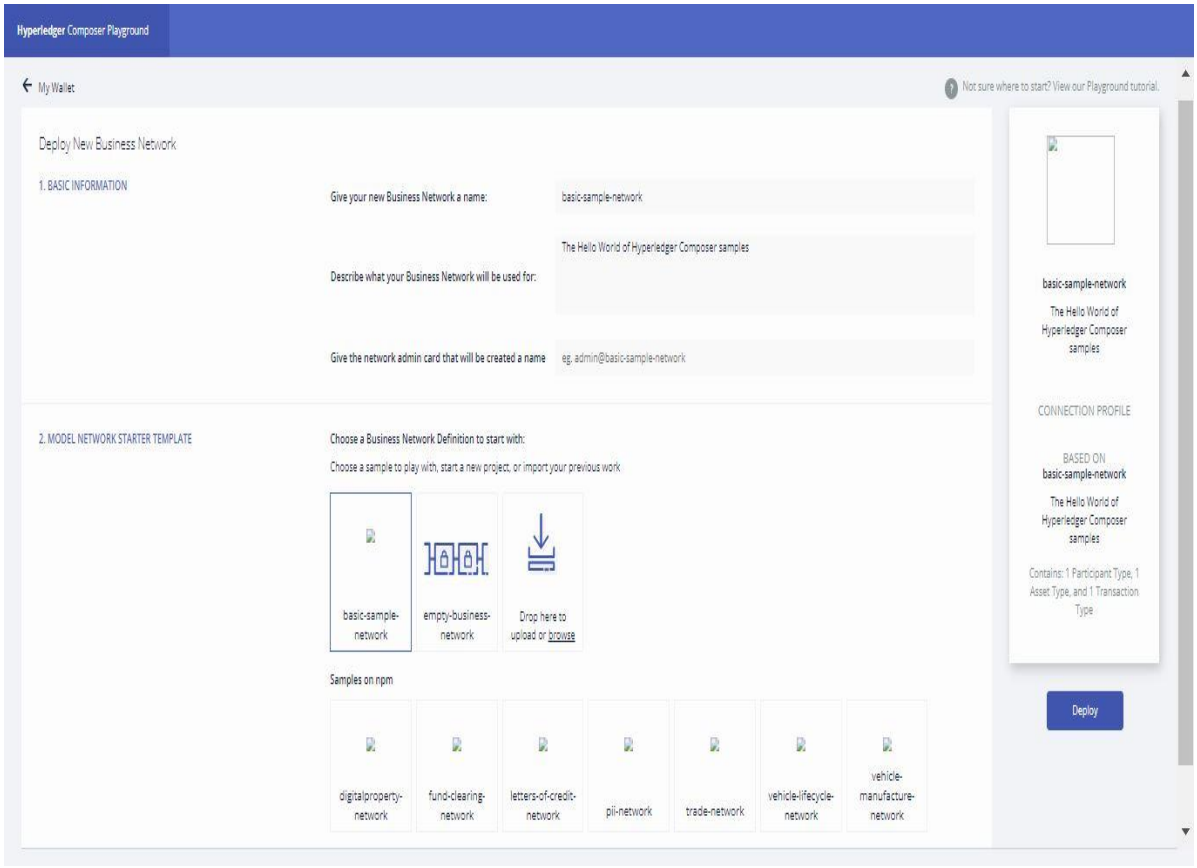


Figure 4.3 : Hyperledger Composer Playground (Layout 3)

After that, we have to choose "Drop here to upload" and then choose our bna file. After that, we must select "Deploy." That is how the Hyperledger playground's business network is set up.

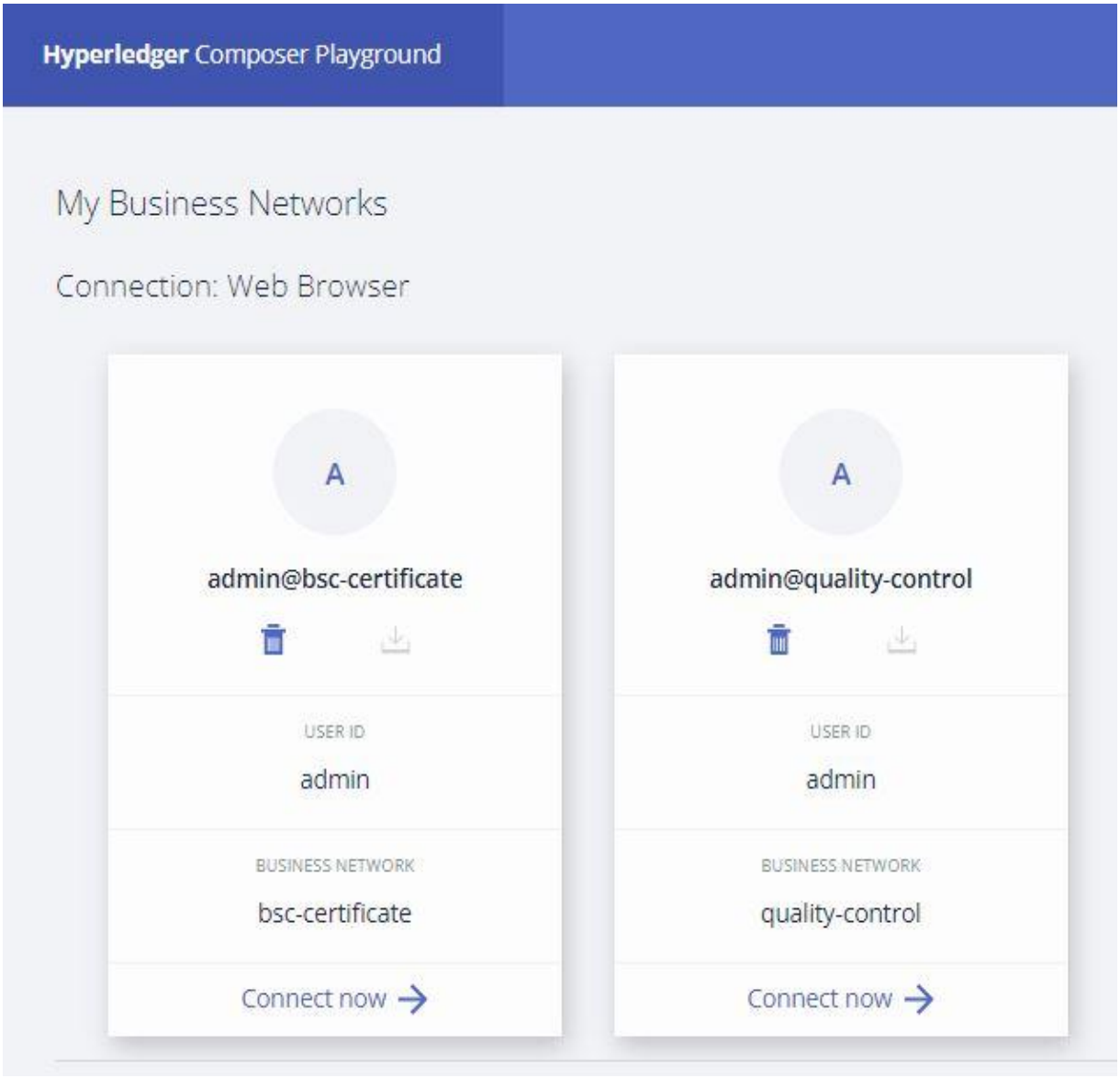


Figure 4.4: Hyperledger Composer Playground (Layout 4)

After clicking " Connect now" we can start our work simulation process.

4.2. Technical Specifications

In this system hyperledger based private blockchain is accomplished. Every asset stored in the blockchain will create a hash which is unique in nature. If anybody changes the hash it will not be matched with the real one. So it's very much difficult to modify anything. The create and updating information will be recorded as a transaction. The entire structure of the system is illustrated below:

A.Creating the assets: Every certificate will be stored as an asset in the hyperledger blockchain. The asset contains the valuable information of certificates such as name, year, cgpa and so on. These assets are collected and accumulated in the asset registry of the private blockchain. It is identified by bsc certificate ID in our system which indicates the particular certificate's circumstantial information and by this id number, it is indexed in the asset registry. Asset registry is nothing but the necessary information collected from the users and stored in the organization asset for further process. However,creating and modifying asset information can only be done by the asset transaction which proves the unchangeable features of blockchain. In this way, history can be preserved.

B.Execution of transactions: Every sustainable operation of our proposed blockchain-based protocol is performed, just like every other blockchain system. Figure-4.5 demonstrates the execution of transactions in this system.To create, access and update the assets this system needs to execute transactions are being used. Once a transaction is placed in one of the chain's blocks, it becomes immutable, which ensures it can't be changed without a massive attack like the majority attack, which is almost impossible to conduct. Whenever an asset is created that means a new certificate is added to the blockchain it will be stored as a transaction. If someone gains access to the certificate, the report will be saved as well. The information transaction will be performed out if it has to be updated. As a result, no one will be able to change the certifications of other people. When a user performs a transaction, the ID is compared to the asset ID. If it isn't matched, it will be taken into account as an illegal or unethical attempt.

Transaction Operation for Certificate Asset Creation, Update, and Emitting event

Implementation of transaction for creating a BSC certificate

```
Input: BSCcertificateId, registrationNo, session, firstName, lastName, age, idNo, fathersName, mothersName,
cgpa, department, universityName, country, date

BSCcertificate Registry ← Entity  of getAssetRegistry('org.BSC.assetTxn.BSCcertificate')
factory ← Instance of  getFactory()
NS ← Instance of org.BSC.assetTxn

BSCcertificate ← Entity  of factory.newResource(NS,'BSCcertificate',BSCdata.BSCcertificateId)

BSCcertificate.owner ← BSCdata.owner
BSCcertificate.registrationNo ← BSCdata.registrationNo
BSCcertificate.session ← BSCdata.session
BSCcertificate.firstName ← BSCdata.firstName
BSCcertificate.lastName ← BSCdata.lastName
BSCcertificate.age ← BSCdata.age
BSCcertificate.idNo ← BSCdata.idNo
BSCcertificate.fathersName ← BSCdata.fathersName
BSCcertificate.mothersName ← BSCdata.mothersName
BSCcertificate.cgpa ← BSCdata.cgpa
BSCcertificate.department ← BSCdata.department
BSCcertificate.universityName ← BSCdata.universityName
BSCcertificate.country ← BSCdata.country
BSCcertificate.date ← BSCdata.date

event ← Entity of factory.newEvent(NS,'BSCcertificateCreated')
    event.BSCcertificateId ← BSCdata.BSCcertificateId
    emit(event)
 return BSCcertificateRegistry.add(BSCcertificate)
```

Implementation of transaction for updating a BSC certificate

```
Input:  BSCcertificateId, newsession, newfirstName, newlastName, newfathersName, newmothersName,
newcgpa

BSCcertificate Registry ← Entity  of getAssetRegistry('org.BSC.assetTxn.BSCcertificate')
factory ← Instance of  getFactory()
NS ← Instance of org.BSC.assetTxn

BSCcertificate.BSCcertificateId ← updatesData.BSCcertificateId
BSCcertificate.session ← updatesData.newsession
BSCcertificate.firstName ← updatesData.newfirstName
BSCcertificate.lastName ← updatesData.newlastName
BSCcertificate.fathersName ← updatesData.newfathersName
BSCcertificate.mothersName ← updatesData.newmothersName
BSCcertificate.cgpa ← updatesData.newcgpa

 event ←  factory.newEvent(NS,'BSCcertificateUpdated')
    event.BSCcertificateId ← updatesData.BSCcertificateId
    emit(event)
   return bC.update(BSCcertificate)
```

Figure 4.5: Transaction Operation for Implementing Creating and Updating Certificate

C.Determining participant types: In order to determine the system's access control, participants are expected in our system.System admin and end user - are the two types of participants to manage the system. The next subsections go over all of the participants' roles.

1) System admin: The system administrator maintains overall control and has access to all of the system's data. Admin has access to the assets and historian records. He can also make adjustments, such as creating it or updating it. Admins have the potential to customize new participants as well as inspect the record. To prevent any form of breach, the system administrator can take appropriate measures.
2) End Users: End users have various restrictions on how they can access the system as a whole. They can only retrieve data related to their certificates but cannot modify it. The ability of end users to create assets is likewise limited.

D.Maintaining Assets in an Asset Registry: All of our system's assets information will be retained in the asset registry which is depicted in Figure-4.6 The asset registry will be adjusted whenever the admin introduces a new asset. Because blockchain has immutable properties, no asset can be truly erased. If a system administrator or end user attempts to access important data about a certain asset, they must input the asset ID into this registry. However,If it is attempted to be deleted, the status will be changed to "deleted". But the asset will remain in place. Furthermore,the updated information will also be stored in the asset registry.

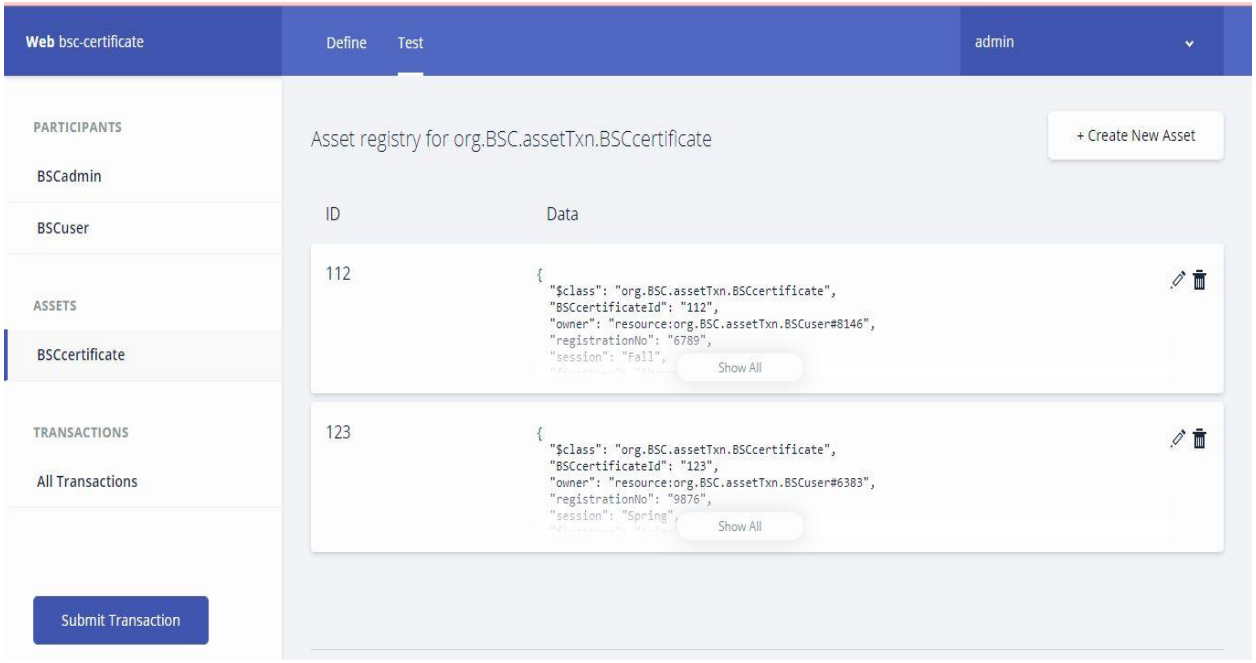


Figure 4.6: Asset Registry for certificate system

E.Transactions from the Historian's Records: In our system, there are primarily two transactions: create transaction and update transaction. Transactions are recorded in blocks in the blockchain system, which are unchangeable. For this feature,the transaction's information can be acquired from the historian record. Historian records are indexed by a particular ID. So in order to retrieve the information, the transaction ID and historian ID is needed. It may be determined whether a transaction exists or has already been confirmed by a user. It may also be verified if there has been an update by using this functionality inside the asset registry and historical record. We can classify the illegal behaviors based on the transaction records. Because each user's certificate id is unique, every attempt to view or retrieve information from another user will result in a transaction which will be recognized in the historian record.

F. Transactions Query in Historian Records: All assets are maintained in the asset registry, and all transactions are preserved in the historian record in our system.Aside from transaction ID, the historical record is additionally indexed by the participant ID accountable for the transaction's

execution and the transaction's type. Whenever we need to access asset information from the asset registry, we use a query.The authentication of Certificate validation by transaction query is illustrated in Figure 4.7. So, a query retrieves transactional information from the historian record.

Validating Certificate authentication by access transaction and access registers

```
Query AccessHistorianRecord:
        description ← "Returns Historian Records"
              statement ← SELECT org.hyperledger.composer.system.
                          HistorianRecord where (transactionType == _$input)
Query AccessAssetRegistry:
              description ← "Returns Asset Registry Elements
              statement ← SELECT org.hyperledger.composer.system.AssetRegistry
                          where(assetID ==$input)
acessSpecificAsset:
                    bnUtil ← Business Network Definition of Certificate System
              assReg ← bnUtil.connection.query('AccessAssetRegistry')
              qry ← bnUtil.connection.buildQuery(assReg)
              return bnUtil.connection.query(qry,{input: assReg#ID)
acessUpdateAssetTransaction:
                     bnUtil ← Business Network Definition of Certificate System
               histRecs ← bnUtil.connection.query('AccessHistorianRecord')
               qry← bnUtil.connection.buildQuery(assReg)
               return bnUtil.connection.query(qry,{input: 'updateCertificateInfo')
checkAssetValidity:
              Input: Asset/BSCcertificateID
              asset ← accessSpecificAsset(assReg#ID)
              if asset IS NOT void
                then historianRecs ← accessUpdateAssetTransaction()
                    for record in historianRecs:
                        if record.BSCcertificateID IS assReg#ID
                        return record
            else
                return VOID
```

Figure 4.7: Authentication of Certificate Validity by Transaction Query

G. Establishing Privacy settings Regulations: To access all of the assets, we needed distinct permissions. Here, we require both permissions to access the asset register and to retrieve information. Whereas, End users do not have permissions to all rights, but administrators do. So, administration has a major effect in establishing privacy setting regulations. Participant instances are kept in the admin composer for securing the certificate system. By using this admin can authenticate users. Users can read and update his/her information by historian record but users can only read the information of the asset registry. Setting permissions on the historian record allows us to control user access which is shown in Figure-4.8. We can also establish permissions for fetching each transaction, such as create and update.

Participant Creation, Assigning Identities and Providing Participants the Access to Execute Transactions

Assigning Identities to End-Use Participant:
participantInstance←composer.participant.add(class:EndUser,participantkey:6789,fname: Shamma, lname: Dilshad, contactNo: 01718421092, email: shammadilshad@gmail.com)
composer.identity.issue(participantInstance,admin@certificate-securing-ledger)

Access Control for System Admin and End Users:
Rule AdminSystemResource:
description← "Grant business network administrators full access to system resources"
participant← "org.hyperledger.composer.system.NetworkAdmin"
operation← ALL
resource← "org.hyperledger.composer.system.**"
action← ALLOW

Rule EndUserPermissionHistorianRecordREAD_CREATE:
description← "Can VIEW and CREATE HistorianRecord inside the Historian"
participant← "org. certificate-securing-ledger.participants.EndUser"
operation← READ, CREATE
resource← "org.hyperledger.composer.system.HistorianRecord"
action← ALLOW

Rule EndUserPermissionAssetRegistryREAD:
description← "Can only VIEW Assets inside AssetRegistry"
participant← "org. certificate-securing-ledger.participants.EndUser"
operation← READ
resource← "org.hyperledger.composer.system.AssetRegistry"
action← ALLOW

Rule EndUserPermissionUpdateCertificateInfoCREATE:
description←"Can EXECUTE UpdateCertificiateInfo transaction"
participant← "org. certificate-securing-ledger.participants.EndUser"
operation← READ
resource← "org. certificate-securing-ledger.certificateInfo. UpdateCertificateInfo"
action← ALLOW

Figure 4.8: Access Control Limiting Blockchain Access

Chapter 5

Result and Discussion

5.1 Experimental Result

We gathered relevant student information from Bangladesh Army International University of Science and Technology (BAIUST) in order to create certificates for our system's inspection. In the following Table 1, a sample of student information is given.

Serial No	ID	Student Name	CGPA	Dept.	Institution
01	1106028	Shamma Dilshad	3.85	CSE	BAIUST
02	1106026	Saima Ahmed	3.78	CSE	BAIUST
03	1205016	Ehsanul Haque	3.90	EEE	BAIUST
04	120529	Zerin Tasnim	3.80	CSE	BAIUST
05	1306013	Fatema Simi	3.70	BBA	BAIUST
06	1306016	Syed Resalat	3.65	BBA	BAIUST

Table 5.1: Arbitrary student Information for system inspection purpose

Admin created assets using the data from Table 1 and then secured the assets in the blockchain as a historian record.

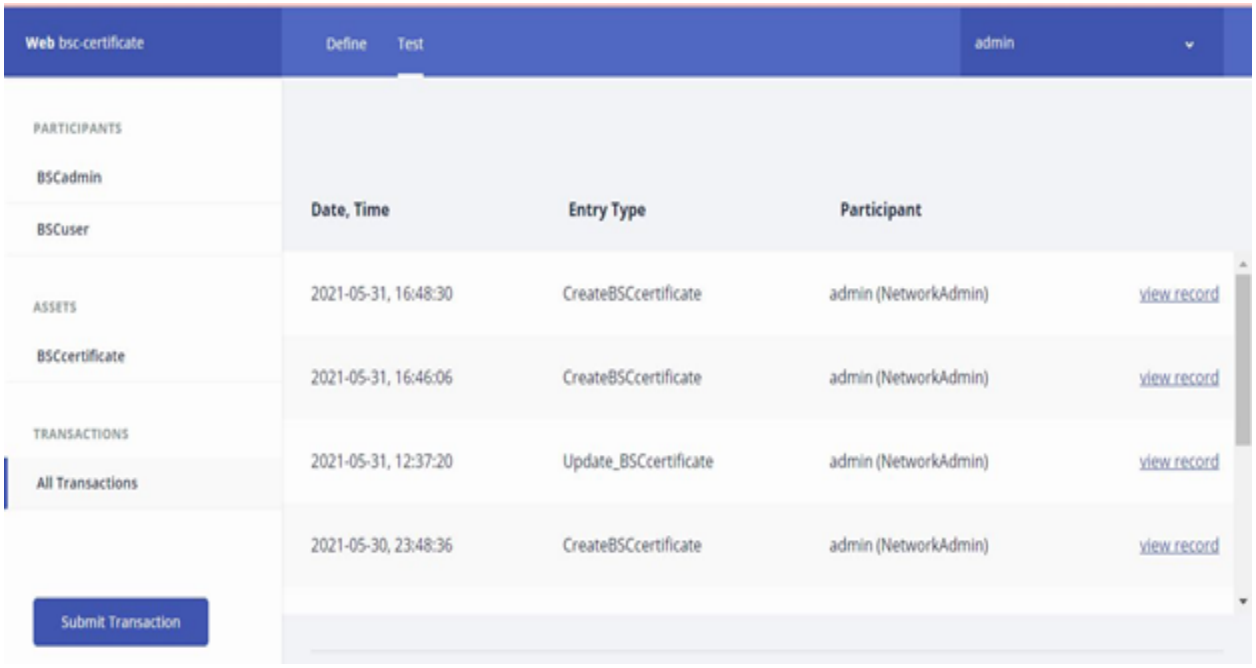


Figure 5.1: Historian Record

Assets are created using the CreateBSCcertificate transaction, and they can also be updated using the Update_BSCcertificate transaction, as shown in the illustration. All transactions show the entry type, participant type of the system shown in figure 5.1.

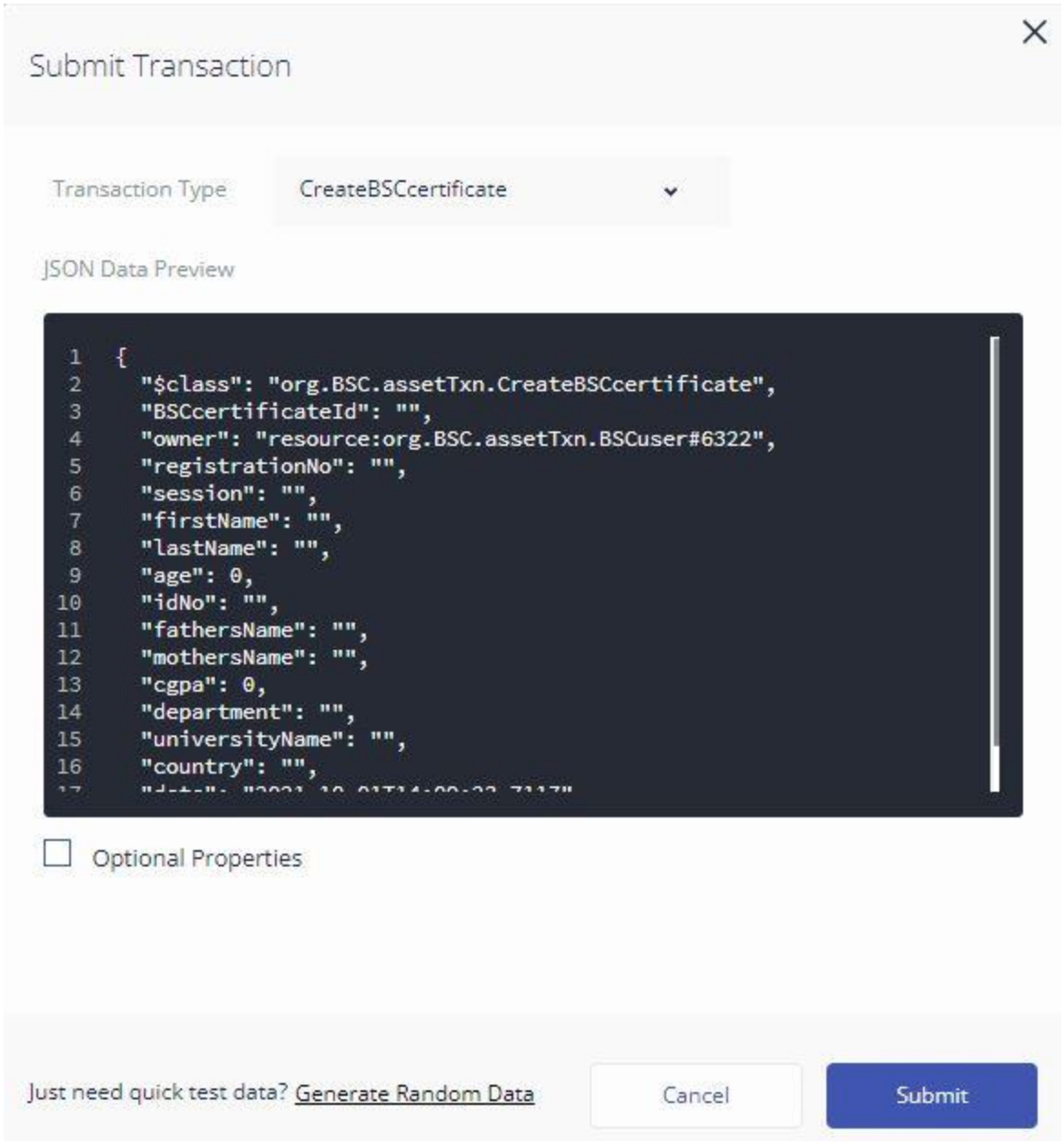


Figure 5.2: Create Asset Transaction

By using CreateBSCcertificate transaction in figure 5.2 we can input unique certificate ID, registration No, session, first name, last name, age , cpa and all the necessary information which are used to create an asset for an user.

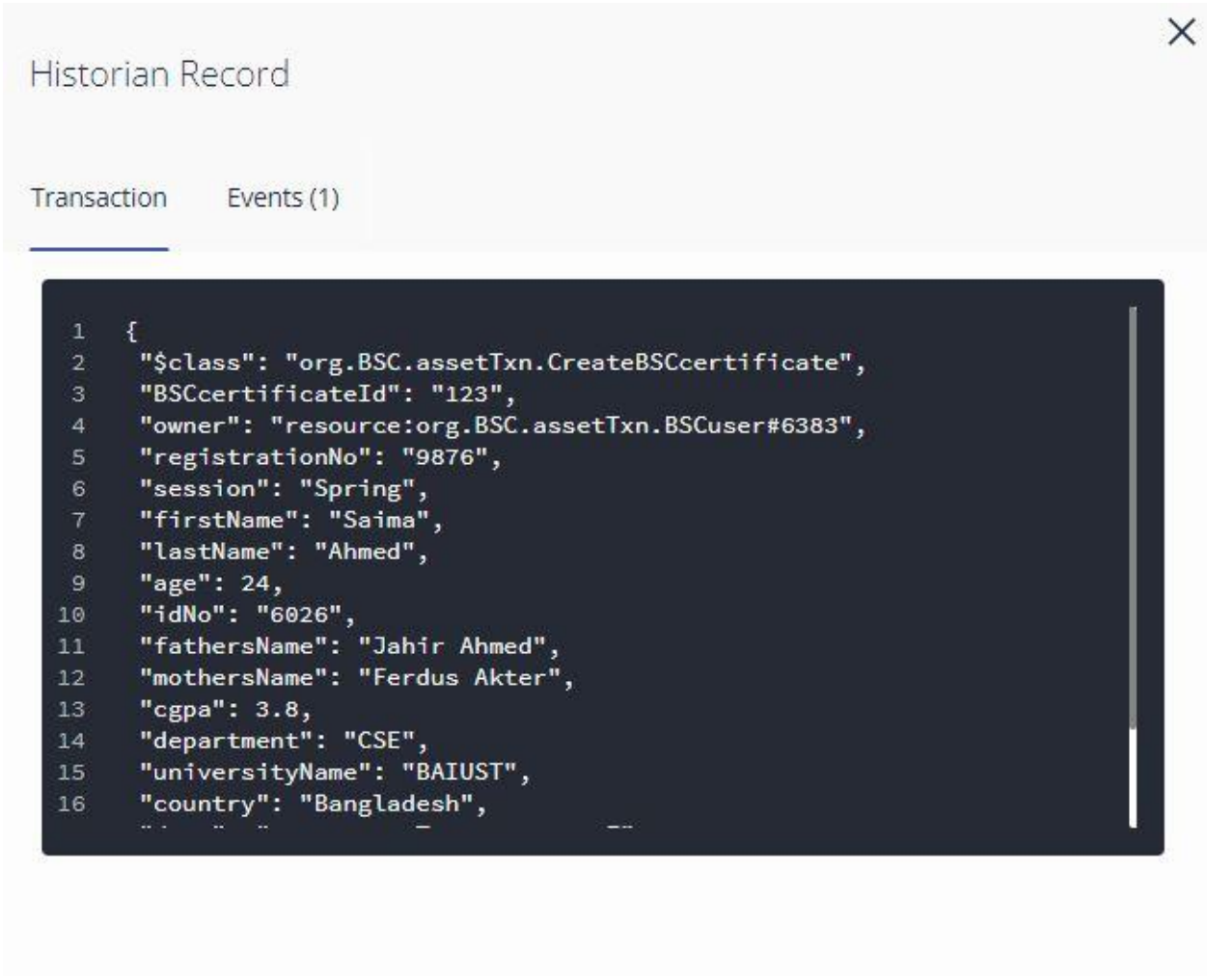


Figure 5.3: Accessing from Historian Record for create asset transaction

In figure 5.3, it is clearly demonstrated in the historian record when anybody wants to access the created asset transaction. There is much information we can get from here, registration no,session,name,age,id no etc. It also illustrates the country name and the university name of a particular user.

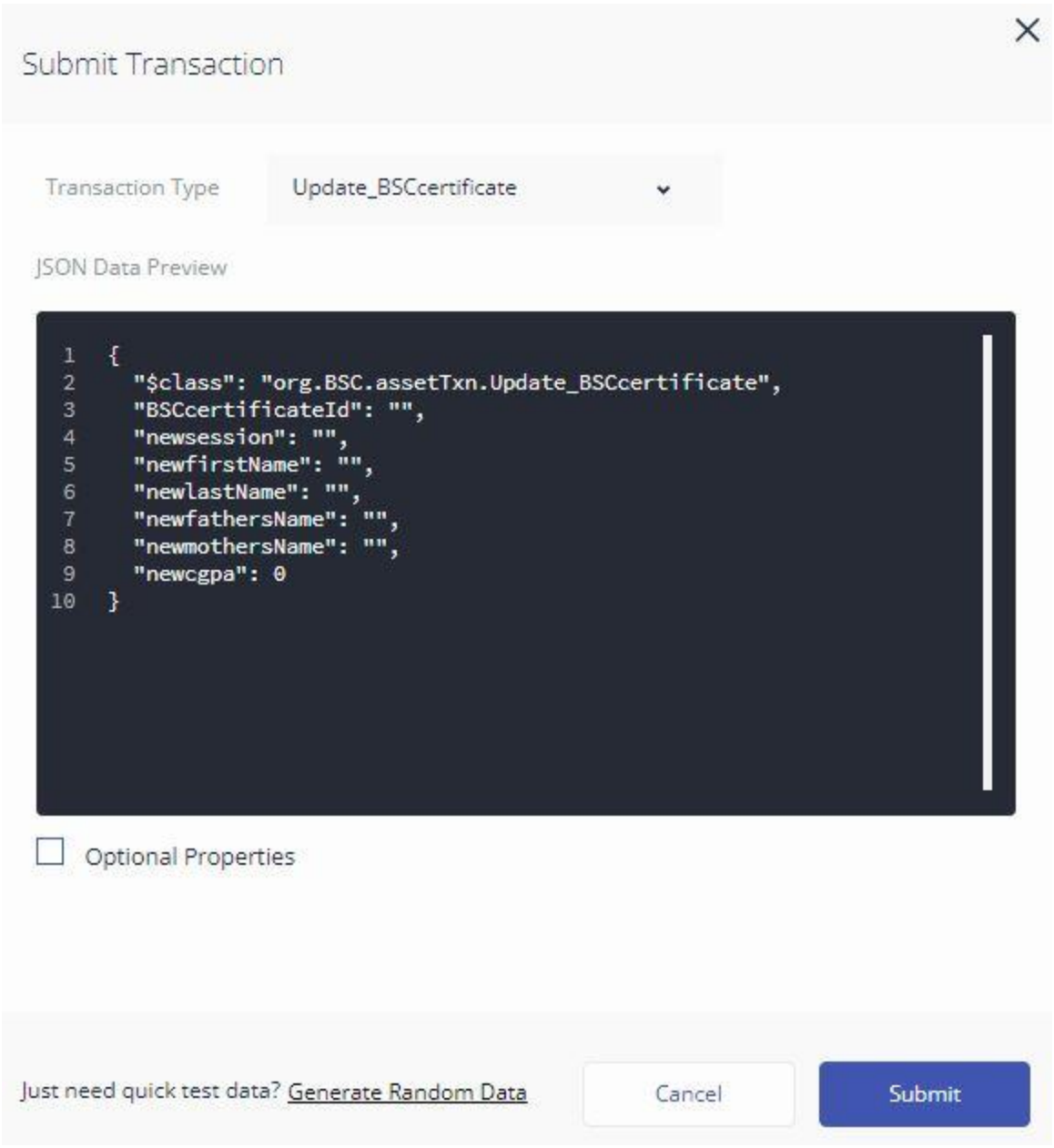


Figure 5.4: Update Asset Transaction

If any update is needed in the certificate system, then an update asset transaction is made which is shown in figure 5.4. It can be of any information except the unique id.

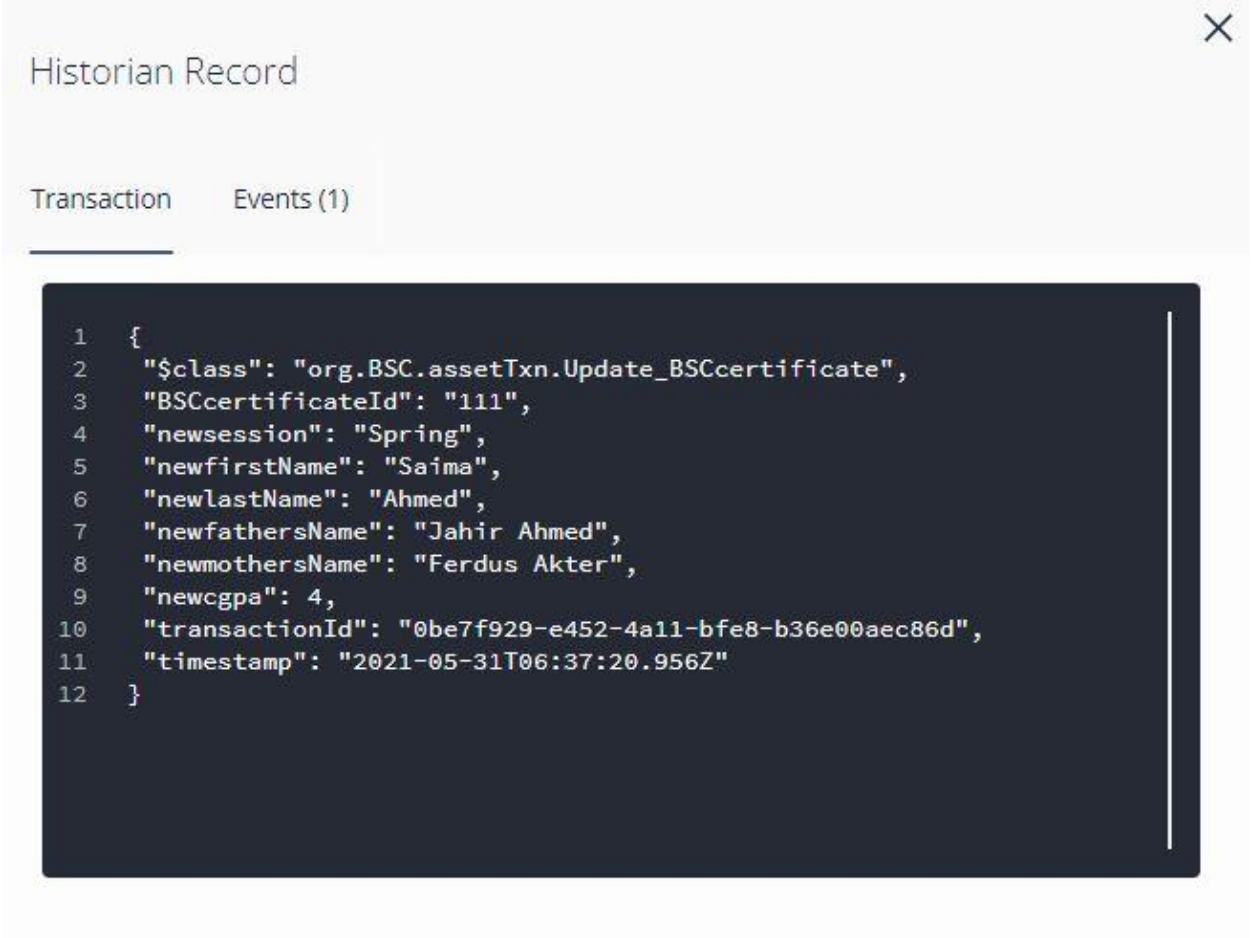


Figure 5.5: Accessing from Historian Record for Update asset transaction

For update access transactions, it can also be viewed in the historian record.New transaction id is also given for demonstrating the changes in information. A time stamp is also given which shows the date and time of the changes.

Number of Transitions	Time for Hyperledger	Time for Ethereum	Time for Bitcoin
350	7.53 seconds	40.9 seconds	42.89 seconds
300	6.43 seconds	32.89 seconds	35.675 seconds
250	4.58 seconds	25.5 seconds	27.68 seconds
200	3.26 seconds	19.85 seconds	21.538 seconds
150	2.45 seconds	12.9 seconds	15.26 seconds
100	1.98 seconds	5.35 seconds	7.185 seconds

Table 5.2: Time Needed to Execute Transactions in Various Blockchain Technologies

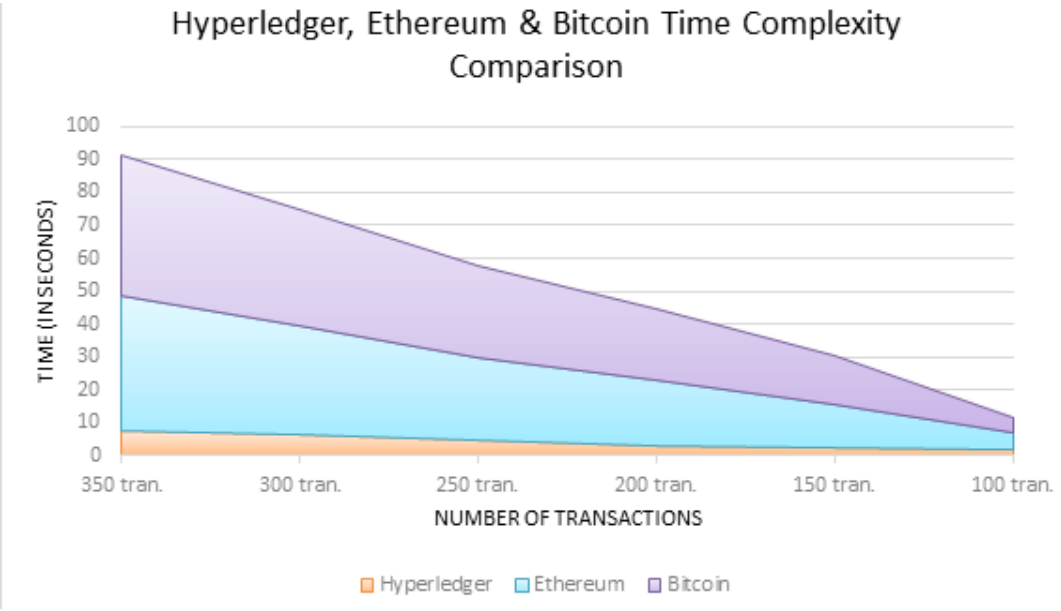


Figure 5.6: Comparison of Computation Time for Three Blockchain Systems

Table 5.2 shows us the time required to execute transactions in various blockchain technologies. We can see that Hyperledger takes less time compared to Ethereum and Bitcoin. In figure 5.6 we can see the time vs number of transactions graph for Ethereum, Bitcoin and Hyperledger which visually demonstrate the computation time for this certificate system.

For estimating the effectiveness of hyperledger in our certificate system,we gathered information from the BAIUST database and checked the differences of required time inserting the information of Bitcoin based blockchain and Ethereum framework. After checking all the things, we came to the conclusion that hyperledger is best for this system.

5.2 Conclusions

Our system is more stable and secure by using hyperledger blockchain. Hyperledger's permissioned and private nature aids the system in executing its operations more reliably. Access to other people's certificates is restricted in the certificate system, which is not feasible in any other permissionless blockchain. Ethereum is not suited for building a certificate system since it requires cryptocurrency to execute any form of transaction and transaction data is generally accessible. In addition, the amount of storage available is minimal. Moreover, the Hyperledger-based authorized network makes use of the decentralized network's historical record, which ensures that a recorded transaction cannot be changed in any way.

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